Cantor's theorem: all countable dense linear orders without endpoints are isomorphic

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-model-theory-basics/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: enumerability is the author's standing countability hypothesis via the !!{enumerable} token; p-isom1's citation carries the [pis] part tag, resolved by label index.

Editorial note. The label thm:cantorQ names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own thm environment in the unit source.

Allowed standard prerequisites (editorial).

- Partial isomorphisms and the back-and-forth property. Same section: definition of partial isomorphism sets and the ISO[p] relation. Lines 466-468 and 513-517: the construction lives entirely inside this vocabulary.
- Back-and-forth yields isomorphism for countable structures. Same part, \ollabel{thm:p-isom1} (source line 992, own card). Cited twice (lines 469 and 519): the abstract transfer from the game property to an actual isomorphism.

Source reference. Original theorem thm-p-isom1: sources/olp-model-theory-basics/pinned-original-source.tex, source lines 963–963. Prerequisite pointer invoked at proof line(s) 499; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Theorem 0.1. *Any two enumerable dense linear orderings without endpoints are isomorphic.*

Proof. Let $\mathfrak{M}_1$ and $\mathfrak{M}_2$ be enumerable dense linear orderings without endpoints, with $<_1 = s{<}M_1$ and $<_2 = s{<}M_2$, and let PIso$I$ be the set of all partial isomorphisms between them. PIso$I$ is not empty since at least $\emptyset \in$ PIso$I$. We

show that PIso$I$ satisfies the Back-and-Forth property. Then $\mathfrak{M}_1 \cong [p]\mathfrak{M}_2$, and the theorem follows by thm-p-isom1 .

To show PIso$I$ satisfies the Forth property, let $p \in$ PIso$I$ and let $p(a_i) = b_i$ for $i = 1, ..., n$, and without loss of generality suppose $a_1 <_1 a_2 <_1 \cdots <_1 a_n$. Given $a \in |\mathfrak{A}|M_1$, find $b \in |\mathfrak{A}|M_2$ as follows:

1. if $a <_1 a_1$ let $b \in |\mathfrak{A}|M_2$ be such that $b <_2 b_1$;
2. if $a_n <_1 a$ let $b \in |\mathfrak{A}|M_2$ be such that $b_n <_2 b$;
3. if $a_i <_1 a <_1 a_{i+1}$ for some i, then let $b \in |\mathfrak{A}|M_2$ be such that $b_i <_2 b <_2 b_{i+1}$.

It is always possible to find a b with the desired property since $\mathfrak{M}_2$ is a dense linear ordering without endpoints. Define $q = p \cup \{\langle a, b\rangle\}$ so that $q \in$ PIso$I$ is the desired extension of p. This establishes the Forth property. The Back property is similar. So $\mathfrak{M}_1 \cong [p]\mathfrak{M}_2$; by thm-p-isom1 , $\mathfrak{M}_1 \cong \mathfrak{M}_2$. □